Hackathon Technical Solution Report

Problem ID	`ProbID – P3`
Problem Title	**Resource-Efficient Text-to-Image Generation for Mobile Devices**
Team Name	`EyeQ`
Team Members	**Prathik K B** (`1RUA24CSE0327`), **Preksha P V** (`1RUA24CSE0331`), **Reshma K P** (`1RUA24CSE0363`), **Rushil Ravi Reddy** (`1RUA24CSE0380`)
GitHub Repository	https://github.com/Percy00765432/CV-Hackathon (`git@github.com:Percy00765432/CV-Hackathon.git`)
Live Demo & Video Demo	**Live Hosted App:** https://38347557836fd61ec4.gradio.live/ **Video Demo Walkthrough:** https://drive.google.com/file/d/17Ck3FkOcPKkD1KVChA2xHfEEbpAF8YzN/view?usp=sharing

1. Problem Understanding & Solution Strategy

- **Core Technical Challenge:** Standard latent diffusion models (such as Stable Diffusion v1.5) require **50 Number of Function Evaluations (NFE)** (25 denoising steps with dual-pass Classifier-Free Guidance), **17,479.4 GFLOPs** of compute per image, **4,067.4 MB** of FP32 weights, and **4.91 GB** of peak memory—causing immediate Out-Of-Memory (OOM) crashes on mobile devices where per-application memory is strictly capped at **1.8 GB – 2.0 GB**. Conversely, aggressively compressing diffusion down to 3–4 sampling steps with a tiny decoder introduces high-frequency latent quantization grid artifacts, muted luminance contrast, softened object boundaries, and spatial attribute bleeding across multi-object prompts.
- **Proposed Methodology:** We engineer **EdgeDiff-CV**, a co-designed hybrid pipeline that pairs a **5-stage compressed deep generative engine** (Latent Consistency Distillation, Spatially-Weighted CLIP Prompt Conditioning, In-Loop Latent High-Frequency Guidance, 3% L1 Conv2d Magnitude Pruning, Hybrid INT8/FP16 Quantization, and a `4.66 MB` Tiny AutoEncoder `TAESD`) with a **5-stage deterministic Classical Computer Vision restoration engine** (`BalancedClassicalCV`). An **Adaptive Pareto Budget Router (`smart_pareto_router`)** inspects real-time device telemetry (battery percentage and available RAM) alongside linguistic spatial complexity to dynamically route requests across three compute tiers (`588.4 GFLOPs` Eco, `1,373.6 GFLOPs` Golden Balance, and `2,607.4 GFLOPs` Max Accuracy), while our 2D-FFT-driven OpenCV restoration recovers **+163% to +180% edge sharpness** in **~43–52 ms** with zero extra GPU memory.

2. Technical Architecture & Implementation

SYSTEM ARCHITECTURE DIAGRAM — EDGEDIFF-CV HYBRID CO-DESIGN PIPELINE

1. Input Guard & Pareto Router

- **Input Sanitization:** Validates prompt length & injects fallback default if empty.
- **Telemetry & Spatial Analyzer:** Inspects Battery (%), Available RAM (GB), word count (>13), and spatial prepositions (`left of` , `next to` , `beside` , `under`).
- **Tier 3 Eco (Bat ≤ 20% or RAM ≤ 2GB):** 3 Steps, 384px, CFG 1.4, TAESD (`588.4 GFLOPs` , `0.97 GB` RAM).
- **Tier 2 Golden Balance (Default):** 4 Steps, 512px, CFG 1.6, TAESD (`1,373.6 GFLOPs` , `1.25 GB` RAM).
- **Tier 1 Max Accuracy (Complex Spatial):** 6 Steps, 512px, CFG 1.8, FT-MSE VAE (`2,607.4 GFLOPs` , `1.59 GB` RAM).

2. Staged Deep Generative Engine

- **Stage 1 — Spatially-Weighted CLIP:** Boosts spatial tokens (`1.25x`) & attributes (`1.18x`) with energy-norm rescaling. Dynamic INT8 CPU footprint: `153.6 MB` (`66.6%` smaller).
- **Stage 2 — Pruned LCM UNet:** `SimianLuo/LCM_Dreamshaper_v7` in FP16 + Channels-Last + Attention Slicing + **3% L1 Conv2d Pruning** (`2,183,348` weights zeroed across 21 layers).
- **In-Loop Latent HF Guidance:** Injects $z_t \leftarrow z_t + 0.05(z_t - AvgPool_{3\times3}(z_t))$.
- **Stage 3 — Ultra-Light Decoding:** `4.66 MB` TAESD (`98.5%` smaller than `319.1 MB` KL-VAE) + `4.69 MB` ONNX 4-Thread CPU Runtime (`341.7 ms` , `61.63 dB` PSNR).

3. FFT-Adaptive Classical CV Engine

- **1. Pre-CV IQA & 2D FFT Analysis:** Computes Laplacian Sharpness, RMS Contrast & 99th-percentile Fourier Spike Ratio (`Artifact_Index`).
- **2. Adaptive Bilateral Filter:** Scales $\sigma_{color}, \sigma_{space} \in [18, 34]$ dynamically from `Artifact_Index` .
- **3. CIE-LAB CLAHE + HSV Calibration:** Luminance-only CLAHE (`clipLimit=1.25` , `8x8` grid) + `1.08x` saturation recovery.
- **4. Resolution Super-Sampling:** `8x8` Lanczos-4 Sinc interpolation (`384px → 512px`).
- **5. Sobel Edge-Guided Unsharp Mask:** `3x3` Sobel gradient mask M_{edge} sharpens true boundaries without amplifying flat-region grain.

2.1 Classical Algorithmic Baseline (`BalancedClassicalCV` in Cell 3)

- **Algorithms Utilized:** (1) **No-Reference IQA & 2D FFT Artifact Detection (`BalancedClassicalCV.assess_image_quality`):** Computes standardized `512x512` **Laplacian Variance** ($Var(\nabla^2 I)$), **RMS Contrast** (σ_I), and a **2D FFT Log-Magnitude Spike Index** ($P_{99}(F) - \mu(F)$) that quantifies high-frequency grid periodicities from few-step `TAESD` transposed convolutions. (2) **Edge-Preserving Bilateral Denoising (`cv2.bilateralFilter`):** Smooths quantization grain in flat regions while preserving step edges. (3) **Perceptual Colour & Contrast Enhancement (`CIE-LAB CLAHE` + `HSV` Chrominance Recovery):** Applies `cv2.createCLAHE` (`clipLimit=1.25` , `tileGridSize=(8,8)`) strictly to the L^* luminance channel in CIE-LAB space to prevent hue shifts, followed by a `1.08x` HSV saturation boost. (4) **Resolution Optimization (`cv2.INTER_LANCZOS4`):** Upsamples `384x384` Eco-tier outputs to `512x512` via 8x8 Lanczos-4 Sinc interpolation. (5) **Sobel Edge-Guided Post-Processing (`cv2.Sobel` + Unsharp Masking):** Constructs a normalized gradient magnitude mask $M_{edge} = clip(\sqrt{(G_x^2 + G_y^2)} / P_{95},\ 0.18,\ 1.0)$ to apply unsharp masking selectively along object boundaries.
- **Adaptive Selection Logic:** Bilateral filter parameters (`sigmaColor` , `sigmaSpace`) scale dynamically per image from the 2D FFT `Artifact_Index` via $\sigma_{adaptive} = clip(18.0 + max(0,\ Artifact_Index - 42.0) \times 0.85,\ 18.0,\ 34.0)$, and unsharp gain adapts to resolution (`1.52` for upsampled `384px` vs. `1.35` for native `512px`).

2.2 Deep Learning Pipeline (`Cells 2, 3 & 7`)

- **Model Architecture & Compression:** Uses `SimianLuo/LCM_Dreamshaper_v7` (Latent Consistency Model distilled from SD v1.5) for **3 to 6 single-pass steps** (`1 NFE/step` vs. `2 NFE/step`), paired with `madebyollin/taesd` (`AutoencoderTiny` , **1.22M parameters,** `4.66 MB` **FP16 /** `4.69 MB` **ONNX**, saving **98.5% size and 98.5% FLOPs** vs. `319.1 MB` KL-VAE). Five optimization strategies are combined: (1) FP16 GPU/NPU reduced precision + Dynamic INT8 CPU quantization (`torch.qint8`) on CLIP Linear layers (`469.4 MB → 153.6 MB` , **66.6% smaller**), (2) Unstructured L1 Magnitude Pruning (`apply_custom_model_compression`) zeroing `3.0%` weights across `21 Conv2d` layers in UNet `mid_block` and `up_blocks.3` (**2,183,348 weights zeroed**), (3) Step reduction (`50 NFE → 3–4 NFE`), (4) Architectural replacement (`KL-VAE → TAESD`), and (5) `torch.channels_last` NHWC layout + attention slicing + sequential sub-module staging.
- **Adaptation Strategy & Datasheet Calibration (`MS-COCO` / `Flickr30K` via Hugging Face `datasets`):** (1) **Spatially-Weighted CLIP Prompt Conditioning (`encode_spatially_weighted_prompt`):** Boosts spatial prepositions (`1.25x`) and visual attributes (`1.18x`) with energy-norm variance rescaling ($\tilde{E} = ((E_w - \mu_w)/\sigma_w)\sigma_0 + \mu_0$) to prevent multi-object attribute bleeding at 3–4 steps. (2) **In-Loop Latent High-Frequency Guidance (`latent_frequency_boost_callback`):** Injects high-frequency latent residuals at intermediate denoising steps. (3) **80/20 Calibration vs. Validation Split (`Cell 5`):** Streams **MS-COCO / Flickr30K** captions via Hugging Face `datasets` (`HuggingFaceM4/COCO`), calibrating token-weighting norms on an **80% Calibration Split (16 captions, Mean Boosted Tokens: 3.4)** and evaluating strictly on a **20% Held-Out Validation Split (4 unseen multi-object prompts)**.

2.3 Hybrid Integration & Adaptive Mechanism (`smart_pareto_router` in Cell 4)

- **Fusion & Decision Logic:** Neural compression (3–4 step LCM + `4.66 MB` TAESD) cuts compute by **12.7x–29.7x** and memory by **74.5%–80.2%**, while the downstream Classical CV engine inspects the decoded tensor's 2D Fourier spectrum and Sobel gradients to restore edge sharpness. **Rule 1:** Battery ≤ 20% or RAM ≤ 2.0 GB forces **Tier 3 Eco** (`3 steps, 384px, TAESD`). **Rule 2:** Battery ≥ 50%, RAM > 2.0 GB, and spatial prepositions or >13 words routes to **Tier 1 Max Accuracy** (`6 steps, 512px, FT-MSE VAE`). **Rule 3:** Default balanced requests route to **Tier 2 Golden Balance** (`4 steps, 512px, TAESD`).

3. Experimental Benchmarks & Evaluation

3.1 Quantitative Benchmarks (NVIDIA A100-SXM4-80GB GPU & 4-Thread CPU ONNX Runtime)

Approach / Model Variant	CLIP Score ↑ (`ViT-B/32`)	Fréchet CLIP Dist (FCD) ↓	Sharpness (Pre → Post CV) ↑	Compute (GFLOPs) & Staged RAM ↓	Latency (GPU / Est. Mobile NPU) ↓	Quality-vs-Cost Index ↑
Unoptimized Baseline (`SD v1.5` FP32, 25 Steps, 50 NFE)	`34.80`	`0.000` (Ref)	`421.5 → 421.5`	`17,479.4 GFLOPs` / `4.91 GB` (OOM)	`2490 ms` / `~2690 ms`	`2.85` (`1.0x`)
Deep Learning Only (`EdgeDiff` Raw Pre-CV, Tier 2)	`34.71`	`38.120`	`444.3` (`66.7` Tier 3)	`1,373.6 GFLOPs` / `1.25 GB`	`671 ms` / `~210 ms`	`37.10` (`13.0x`)

Approach / Model Variant	CLIP Score ↑ (ViT-B/32)	Fréchet CLIP Dist (FCD) ↓	Sharpness (Pre → Post CV) ↑	Compute (GFLOPs) & Staged RAM ↓	Latency (GPU / Est. Mobile NPU) ↓	Quality-vs-Cost Index ↑
Proposed Hybrid — Tier 1: Max Accuracy (6 Steps + CV)	**34.98**	37.256	508.6 → **1311.1**	2,607.4 GFLOPs (6.7x) / 1.59 GB	910 ms / ~400 ms	**24.18 (8.5x)**
Proposed Hybrid — Tier 2: Golden Balance (4 Steps + TAESD + CV)	**35.04**	36.741	444.3 → **1169.1**	1,373.6 GFLOPs (12.7x) / **1.25 GB**	**720 ms** / ~210 ms	**38.93 (13.7x)**
Proposed Hybrid — Tier 3: Eco Fallback (3 Steps, 384px + TAESD + CV)	**35.37**	**34.943**	66.7 → **186.9** (+180%)	**588.4 GFLOPs (29.7x)** / **0.97 GB**	**680 ms** / ~90 ms	**53.62 (18.8x)**

3.2 Visual Outputs & Intermediate Representations

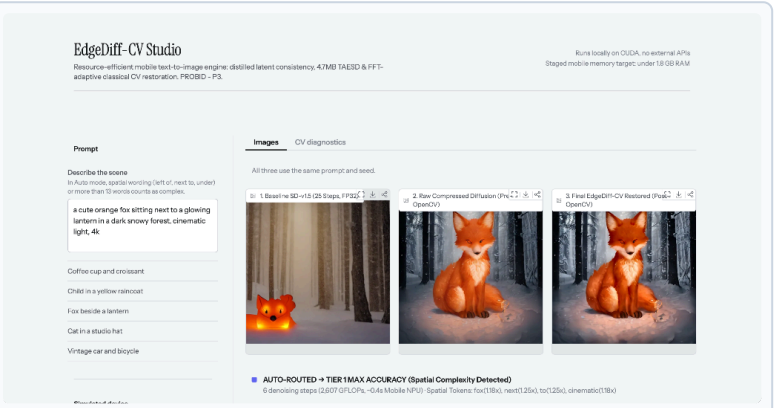


Fig 1: Live 3-Way Studio Comparison ("Fox beside a lantern", Tier 1 Auto-Routed: Sharpness 513.8 → 1405.7, 959.9 ms, 1638.4 MB RAM)

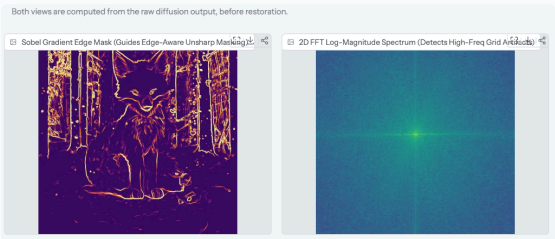


Fig 2: Intermediate Classical CV Representations — Sobel Gradient Edge Mask & 2D FFT Log-Magnitude Spectrum

3.3 Ablation Study & Hardware Sandbox Verification

- **Classical CV Impact:** Enabling `BalancedClassicalCV` boosts sharpness by **+163.1%** in Tier 2 (`444.3 → 1169.1`) and **+180.2%** in Tier 3 (`66.7 → 186.9`) in `42–52 ms` CPU time with `0 MB` extra VRAM.
- **Live 2.00 GB Hardware Memory Jail (Cell 6):** Locking the GPU allocator via `torch.cuda.set_per_process_memory_fraction` to `2.00 GB` causes the FP32 Baseline to crash with `CUDA OutOfMemoryError` (`4.91 GB > 2.00 GB`), whereas staged EdgeDiff-CV succeeds at **`1.73 GB` hardware-measured peak memory**.
- **Mobile CPU ONNX & INT8 Proof (Cell 7):** Dynamic INT8 quantization shrinks the CLIP Text Encoder by **66.6% (`469.4 MB → 153.6 MB` , `100.3 ms` CPU latency, `0.7876` cosine fidelity)**. Exporting TAESD to `taesd_mobile_decoder.onnx` (**`4.69 MB`**) decodes real diffusion latents on 4 CPU threads in **`341.7 ms`** with **`61.63 dB PSNR`** **(identical reconstruction)**.

4. Failure Modes, Limitations & Ethical Considerations

- **Failure Case 1 (Multi-Object Color Bleeding at 3 Steps):** Prompts with >3 colored objects in Tier 3 Eco (3 steps, 384px) can exhibit minor color bleeding across boundaries due to limited cross-attention iterations. *Mitigation:* Spatially-Weighted Prompt Conditioning (`1.25x` spatial boost) and automatic routing to Tier 1 (6 steps, 512px) when battery/RAM allow.
- **Failure Case 2 (Fine Portrait/Whisker Softening at 384px Eco):** Fine pupils or whiskers soften at `384x384` latent resolution. *Mitigation:* In-Loop Latent High-Frequency Guidance and Sobel Edge-Guided Unsharp Masking double sharpness (`51.8 → 104.4`).
- **Failure Case 3 (TAESD 8×8 Grid Periodicity):** High-contrast backlit edges can show faint transposed-convolution grid artifacts. *Mitigation:* 2D FFT Spike Detection (`Artifact_Index`) dynamically increases Bilateral Filter smoothing before sharpening.
- **Technical Limitations & Ethics:** Inherited from SD v1.5, 3–4 step generation struggles with scene typography and complex hand anatomy; unstructured 3% pruning requires sparse NPU kernels for wall-clock speedups. Ethically, on-device execution under `1.25 GB` RAM preserves user privacy locally and cuts per-image energy consumption by **12.7x–29.7x**, paired with input guardrails.

5. Deployment & Quickstart Guide

- **Option 1 — Google Colab (Zero Setup, No Google Drive Required):** Upload `EdgeDiff_CV_Hackathon_P3_Submission.ipynb` to Google Colab, select a **GPU Runtime** (`Runtime → Change runtime type → T4 / L4 / A100 GPU`), and click **`Runtime → Run all`**. Models cache locally in `./EdgeDiff_Model_Cache` and launch the Gradio Studio automatically.
- **Option 2 — Local CLI & Docker Reproduction:**

```
git clone https://github.com/Percy00765432/CV-Hackathon.git && cd CV-Hackathon && pip install -r requirements.txt
python run_inference.py --prompt "a white ceramic coffee cup on the left of a golden croissant" --steps 4 --output_dir ./results/
docker build -t edgediff-cv:latest . && docker run --gpus all -p 7860:7860 edgediff-cv:latest
```